Avery Hughes

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EDUCATION

Bachelor of Science in Cybersecurity, Minor in Computer Science | Indiana Institute of Technology May 2027

WORK EXPERIENCE

Program Developer & Digital Literacy Educator | Ball Venture Grant | Fort Wayne, IN October 2025 - Current

- Developed and delivered hybrid training sessions for seniors on technology basics and cybersecurity awareness.
- Built an AI-powered platform to personalize learning, track progress, and provide ongoing digital support.
- Increased participants in digital confidence and engagement by teaching safe online practices and social connectivity tools.

IT Service Agent | Indiana Institute of Technology | Fort Wayne, IN September 2025 - Current

- Resolve 100+ IT tickets monthly, troubleshooting network, software, MFA, SSO, Microsoft Office, and hardware issues.
- Assist in deploying and maintaining campus-wide IT infrastructure for 1,500+ students and faculty.
- Reduce system downtime by 30% through monitoring and rapid response.

Data Center Technician | Indiana Institute of Technology | Fort Wayne, IN August 2025 – Current

- Redesigned data center layout with Proxmox and ZFS/LVM-thin to improve efficiency, airflow, and system reliability.
- Integrated NetBox IPAM, Splunk monitoring, and LDAP to centralize visibility and strengthen infrastructure management.
- Standardized configurations and documentation to enhance maintainability and reduce operational overhead.

Undergrad Research Assistant | University of Notre Dame | Remote February 2025 - Current

- Build decentralized backend APIs using Python, FastAPI, Solidity, and IPFS to support Web3DB's distributed data architecture.
- Designed and implemented encryption, key-management, and access-control modules to enhance security and data integrity.
- First-author publication in *Springer CCIS* — "Secure and Scalable Data Management Using Web3DB."

Indiana Tech Cyber Warriors | Indiana Institute of Technology | Fort Wayne, IN July 2024 - Current

Team Lieutenant | Unix Specialty Lead | Offensive Security Co-Lead

- Lead Unix, offensive security, and team-wide training sessions for a 20-member competitive cybersecurity team.
- Develop exercises, documentation, and reports focused on penetration testing, system hardening, and incident response.
- Coordinate competition strategy and execution, contributing to high team placements across cybersecurity events.

COMPETITIONS

- National Collegiate Cyber Defense Competition | 2x Indiana Qualifier | 2x Midwest Qualifier | 1x National Appearance
- Department of Energy CyberForce Competition | 1x Appearance
- Collegiate Penetration Testing Competition | 2x Appearance
- University At Buffalo Lockdown | 1x Appearance

PROJECTS

Core Infrastructure Dockerization

- Containerized core services with Docker & Nginx reverse proxy (SSL/TLS, DNS) for secure, scalable access.
- Implemented version-controlled configs and automated deployments to streamline updates and maintenance.

Local AI Deployment

- Deployed AI/ML models on internal Linux infrastructure for secure research, reducing reliance on cloud services.
- Implemented GPU passthrough and containerized environments for reproducible training and inference.

COAL | Cyber Defense Competition Script

- Built SSH automation tool to mass-execute scripts across hosts; deployed in National CCDC and UBUFF Lockdown.
- Improved team response time by automating repetitive tasks and standardizing configurations during competitions.

magicLAMP | Automated LAMP Stack Deployment

- Authored Bash script to provision Linux, Apache, MySQL, PHP stacks, cutting setup time from hours to minutes.
- Enhanced portability by designing idempotent installs with consistent, tested configurations.

For a full list of projects, please visit my website

Skills, Leadership, Honors & Awards

- **Skills**: Python, Bash, Solidity, FastAPI, Docker, Nginx, IPFS, Proxmox, ZFS/LVM, Active Directory, DNS/DHCP, LDAP, Splunk
- **Security**: Penetration Testing, Incident Response, System Hardening, Offensive Security, Network Security
- **Leadership:** Team Lieutenant (Cyber Warriors), Unix Lead, Offensive Security Co-Lead, Technical Training & Documentation
- **Honors:** Springer CCIS Author, CCDC National Appearance, Ball Venture Grant Recipient, Dean's List